

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement category	Requirement description	Priority
1	User Authentication	The system shall allow authorized bank employees to access the application using secure login credentials.	High
2	Access Control	Different user roles shall be supported to ensure that only authorized personnel can view, manage, or modify applicant information and prediction results.	High
3	System Integration	The application shall communicate with the trained XGBoost prediction model and support optional cloud deployment services when required.	High
4	Loan Prediction Processing	The system shall accept applicant information, validate the entered data, execute the prediction model, and store the generated results for future reference.	High
5	Reports and History	Users shall be able to view loan prediction reports and access previously generated prediction records.	Medium
6	Decision Logic	Loan approval recommendations shall be generated based on applicant information including income, education, employment status, loan amount, property area, and credit history.	High
7	Data Security Compliance	The application shall protect applicant information by following secure banking data handling practices and applicable privacy regulations.	High
8	Model Administration	Administrators shall be able to replace or retrain the prediction model without changing the application's user interface.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Processing Speed	The system should generate loan prediction results immediately after receiving applicant details.	Response time below 2 seconds
2	Scalability	The application should support a growing number of users and loan applications without affecting performance.	Minimum 500 concurrent users
3	Security	Sensitive applicant information should be protected using secure communication channels and encrypted storage mechanisms.	HTTPS enabled with secure data protection
4	Availability	The application should remain accessible and operational with minimal downtime.	99.9% availability
5	User Experience	The interface should be simple, responsive, and compatible across commonly used web browsers.	Responsive UI with cross-browser compatibility
6	Maintainability	The software should follow a modular architecture so that future enhancements and model updates can be implemented easily.	Modular and reusable code structure